

Experience

Full Stack + AI Engineering Intern — PropertySearchGPT — Toronto, ON Jan 2026 - Apr 2026

- Designed a two-tier LangGraph agent pipeline - a nano LLM classifies intent and short-circuits simple queries; a tool-calling model handles geo/listing requests, cutting inference cost on low-complexity traffic
- Built a property price estimator ensembling LightGBM, XGBoost, and CatBoost with Optuna HPO (50 trials, 5-fold CV, temporal split) and custom comparable scoring with exponential decay weighting by distance and recency
- Designed a dynamic ML + comparable alpha-blending algorithm that caps ML weight at 30% when high-quality nearby comps exist, with confidence scoring derived from MedAPE and comp proximity
- Implemented a two-stage prompt injection guardrail: regex pre-filter over 15 attack patterns (zero-cost fast path), followed by an LLM classifier that fails open on outage to avoid blocking legitimate users

Machine Learning Research Assistant — CAMJ, University of Waterloo — Waterloo, ON May 2025 - Aug 2025

- Developed supervised ML classifiers (Random Forest, XGBoost) to predict weld quality from real-time process data, leveraging stratified k-fold cross-validation, Nested Cross-Validation, and class weighting to address data imbalance; Achieved 97% accuracy, 95% recall, and 96% F1-score for predictive performance on unseen data
- Fine-tuned YOLOv11 with data augmentation to detect weld defects from etched cross-sections, achieving mAP@0.5 of 0.89; integrated OpenCV contour analysis to localize and visualize defect boundaries, improving recall by 40%

Battery Hardware Developer — Midnight Sun — Waterloo, ON Sep 2024 - Apr 2025

- Executed cell configuration analysis to optimize energy density, increasing the total battery pack capacity to 5.25 kWh
- Implemented a plated cell-level fusing system to establish a physical barrier, successfully mitigating thermal runaway risk

Piano Tutor — Toronto, ON Sep 2023 - Aug 2024

Projects

Visual Music Classifier — Python, PyTorch, CLIP, CLAP, FAISS, FastAPI, UMAP, torchaudio, SQLite

- Trained dual 512→256 projection heads with InfoNCE contrastive loss to align frozen CLIP image and CLAP audio encoders into a shared embedding space; optimized with AdamW and cosine annealing LR
- Curated a ~2,740-track dataset via iTunes Search API with 60+ targeted genre/aesthetic queries and deduplication by artist + title; built incremental embedding extraction to expand the dataset without full reprocessing
- Indexed projected 256-dim embeddings in FAISS for real-time image-to-music retrieval; applied UMAP (cosine metric) for embedding space visualization with score-weighted position interpolation to place query images on the map at inference

Autonomous LiDAR Navigation for Mobile Robot — C++, Docker, ROS2, Foxglove, Ubuntu, LiDAR

- Developed ROS2 nodes on Ubuntu to process LiDAR point clouds into a 2D costmap for real-time obstacle detection
- Fused costmap and odometry data to generate a dynamic world model representing the robot's driving environment
- Implemented A* path planning algorithm to generate optimized, collision-free routes through complex, mapped terrain
- Applied Pure Pursuit control for smooth differential drive navigation, enabling sub-centimeter path tracking accuracy

FTC Competitive Robot — OpenCV, TensorFlow, OnShape, Mechanical Design, 3D Printing

- Engineered a motor-controlled Musumi Linear Slider robotic arm, with object recognition in OpenCV and Tensorflow

Skills

- **Programming & Scripting:** Python, C, C++, Java, TypeScript, JavaScript, Bash, Linux, Ubuntu
- **ML/AI & Data Science:** PyTorch, TensorFlow, scikit-learn, XGBoost, OpenCV, FAISS, LightGBM, Hugging Face, Optuna
- **Web & API Development:** Next.js, React, FastAPI, LangChain, LangGraph, REST APIs, Prisma, Gemini API
- **DevOps & Tools:** Git, Docker, CMake, ROS2, SSH, Foxglove, GitHub, Vercel, MongoDB, SQLite, torchaudio

Education

University of Waterloo — BAsC, Mechatronics Engineering

June 2029